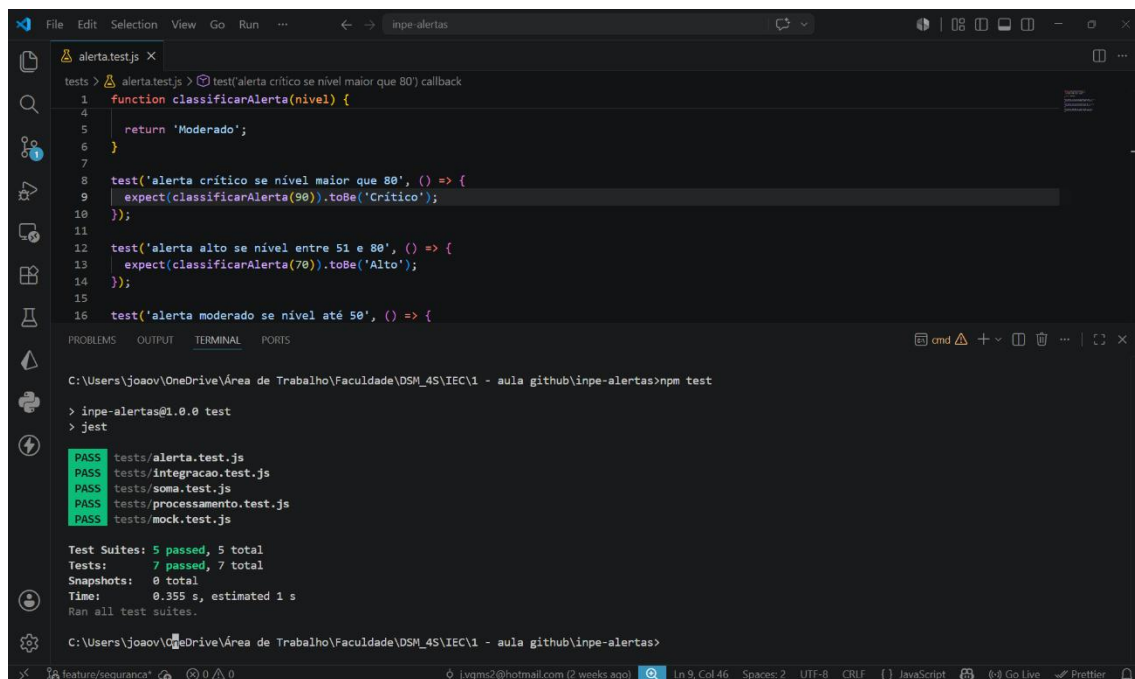
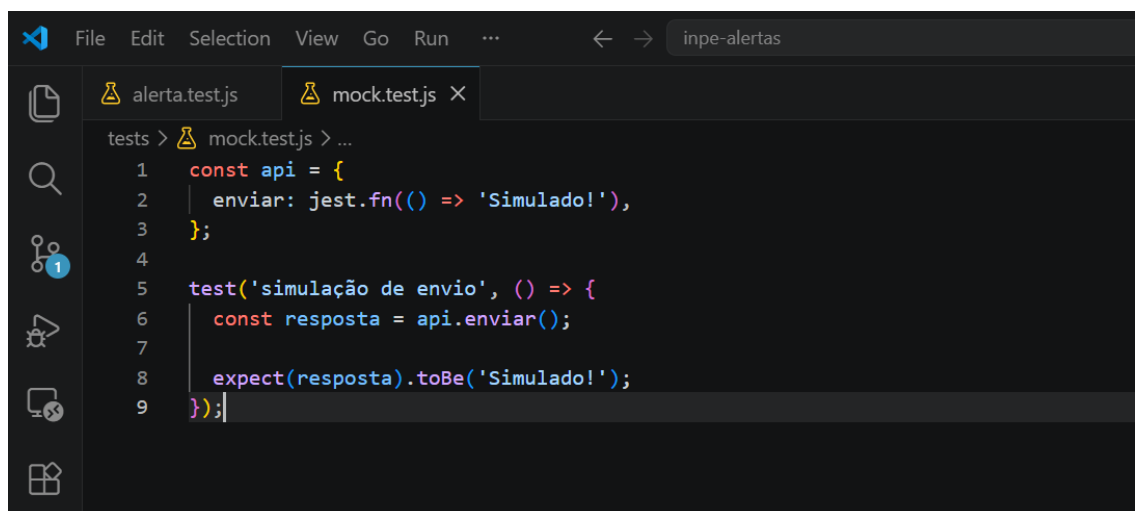


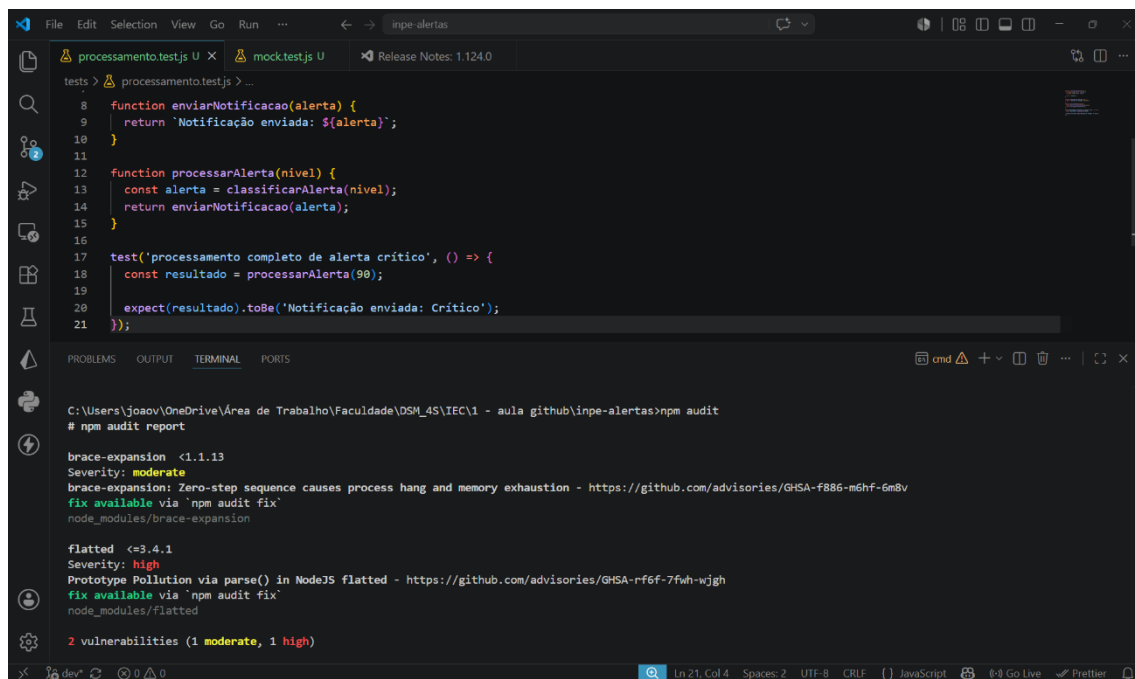
Exercicio 1



Exercicio 2



Exercicio 3



The screenshot shows a VS Code editor with a file named `processamento.test.js`. The code contains two functions: `enviarNotificacao` and `processarAlerta`, along with a Jest test. The terminal window shows the output of `npm audit`, which identifies two vulnerabilities: `brace-expansion` (moderate severity) and `flatted` (high severity). Both are listed as fixable via `npm audit fix`.

```
function enviarNotificacao(alerta) {
  return `Notificação enviada: ${alerta}`;
}

function processarAlerta(nivel) {
  const alerta = classificarAlerta(nivel);
  return enviarNotificacao(alerta);
}

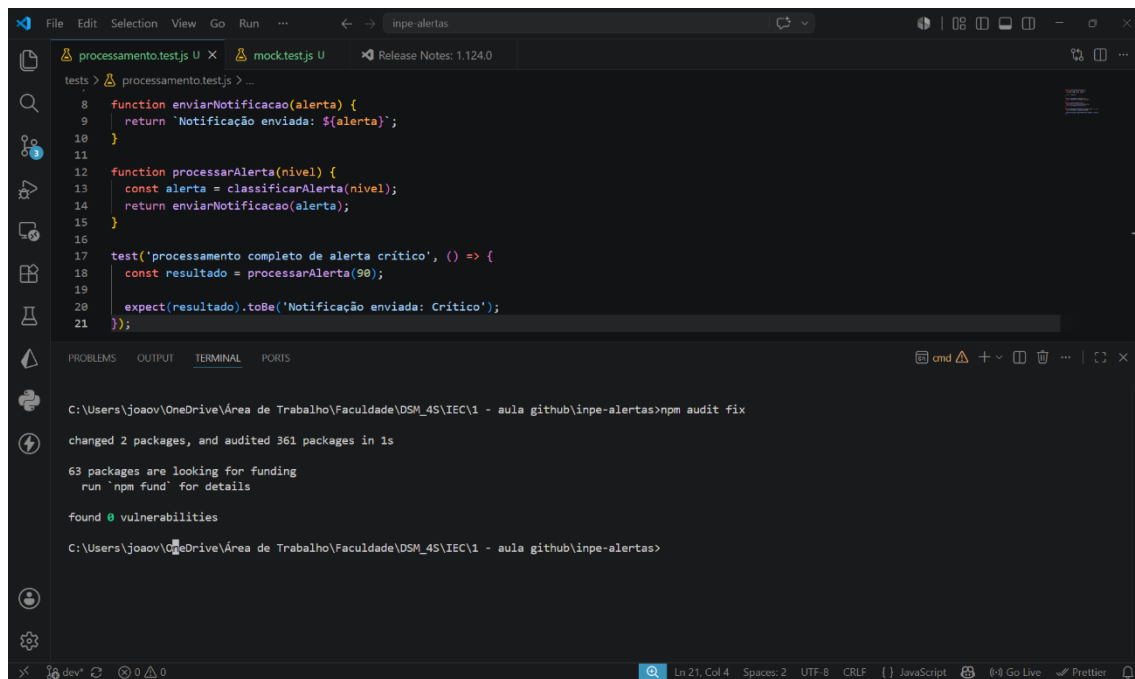
test('processamento completo de alerta critico', () => {
  const resultado = processarAlerta(90);
  expect(resultado).toBe('Notificação enviada: Critico');
});
```

```
C:\Users\joaov\OneDrive\Área de Trabalho\Faculdade\DSM_4S\IEC\1 - aula github\inpe-alertas>npm audit
# npm audit report

brace-expansion <1.1.13
Severity: moderate
brace-expansion: Zero-step sequence causes process hang and memory exhaustion - https://github.com/advisories/GHSA-f886-m6hf-6m8v
fix available via `npm audit fix`
node_modules/brace-expansion

flatted <=3.4.1
Severity: high
Prototype Pollution via parse() in NodeJS flatted - https://github.com/advisories/GHSA-rf6f-7fwh-wjgh
fix available via `npm audit fix`
node_modules/flatted

2 vulnerabilities (1 moderate, 1 high)
```



The screenshot shows the same VS Code editor after running `npm audit fix`. The code in `processamento.test.js` is identical to the previous screenshot. The terminal window now shows the output of `npm audit fix`, which reports that 2 packages were changed and 361 packages were audited in 1s. It also shows that 63 packages are looking for funding and that no vulnerabilities were found.

```
C:\Users\joaov\OneDrive\Área de Trabalho\Faculdade\DSM_4S\IEC\1 - aula github\inpe-alertas>npm audit fix

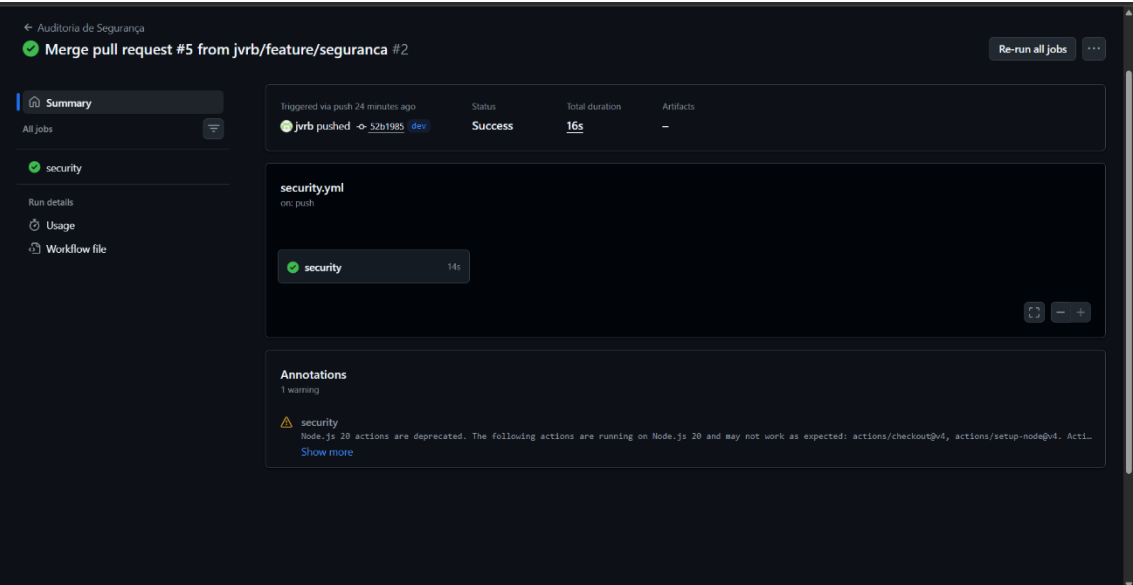
changed 2 packages, and audited 361 packages in 1s

63 packages are looking for funding
  run `npm fund` for details

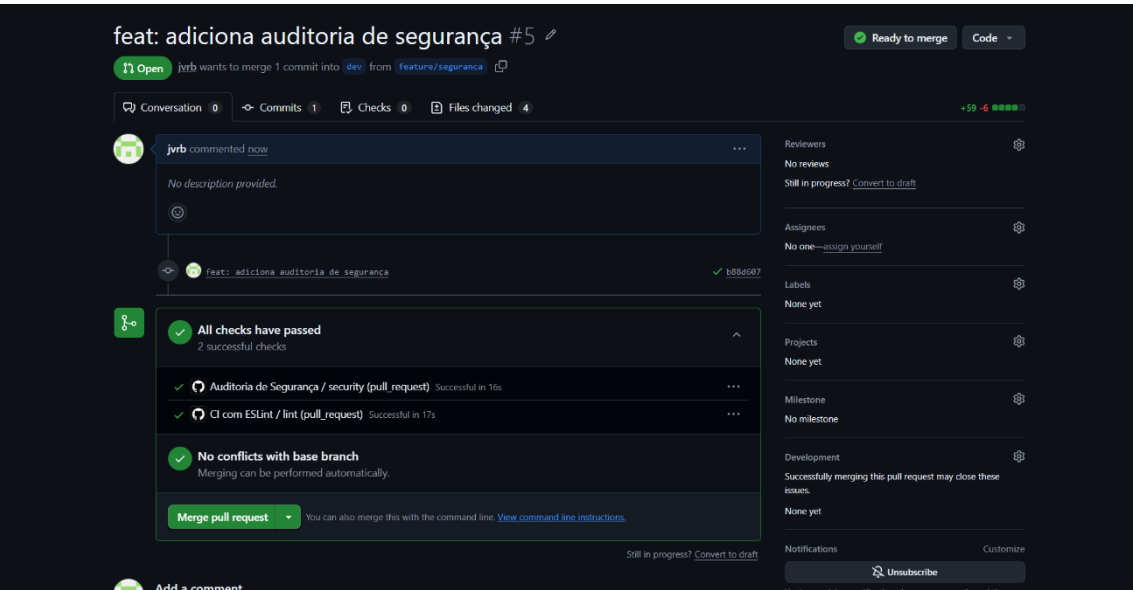
found 0 vulnerabilities

C:\Users\joaov\OneDrive\Área de Trabalho\Faculdade\DSM_4S\IEC\1 - aula github\inpe-alertas>
```

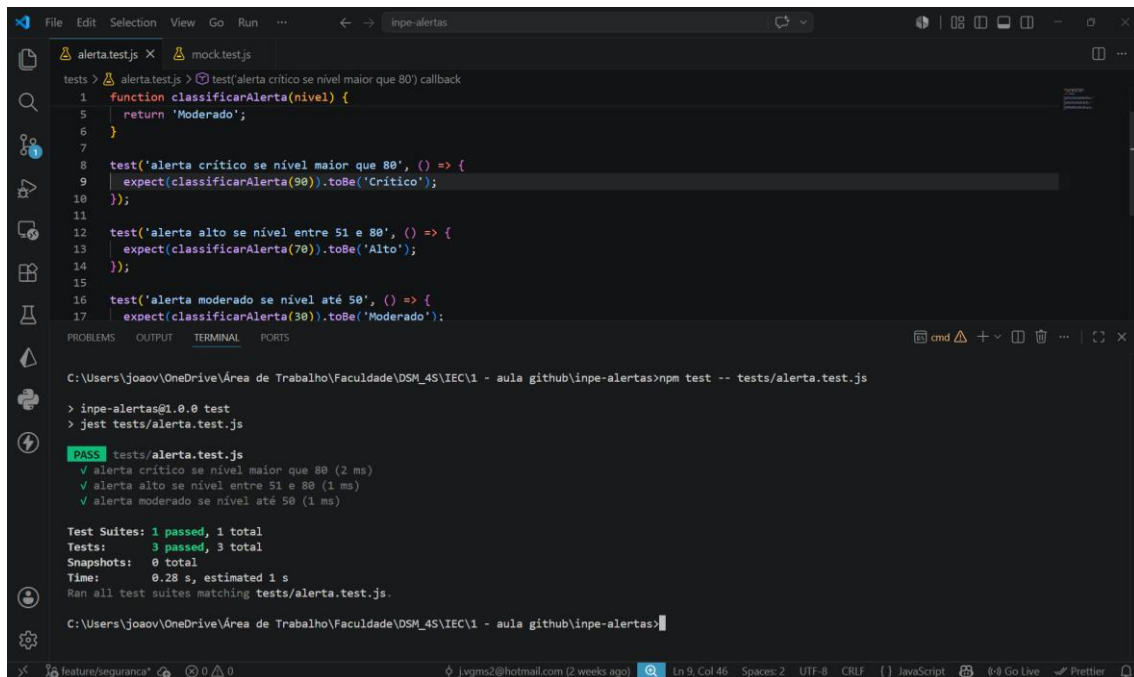
Exercicio 04



Exercicio 05



Exercicio 06



The screenshot shows the VS Code editor with a file named `alerta.test.js` open. The code defines a `classificarAlerta(nivel)` function and three Jest tests. The terminal at the bottom shows the command `npm test -- tests/alerta.test.js` being executed, resulting in three passed tests. The status bar at the bottom indicates the file is `feature/seguranca` and the user is `jvgms2@hotmail.com`.

```
tests > alerta.test.js > test('alerta critico se nivel maior que 80') callback
1 function classificarAlerta(nivel) {
5   return 'Moderado';
6 }
7
8 test('alerta critico se nivel maior que 80', () => {
9   expect(classificarAlerta(90)).toBe('Critico');
10 });
11
12 test('alerta alto se nivel entre 51 e 80', () => {
13   expect(classificarAlerta(70)).toBe('Alto');
14 });
15
16 test('alerta moderado se nivel até 50', () => {
17   expect(classificarAlerta(30)).toBe('Moderado');
```

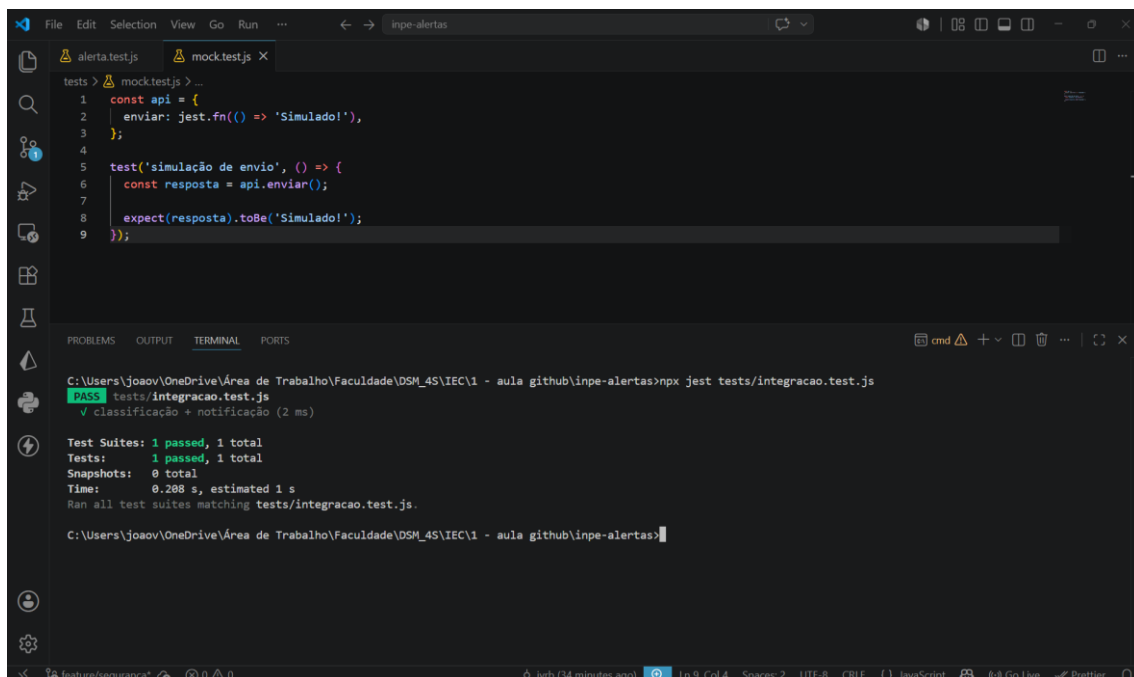
```
C:\Users\joaov\OneDrive\Área de Trabalho\Faculdade\DSM_4S\IEC\1 - aula github\inpe-alertas>npm test -- tests/alerta.test.js
> inpe-alertas@1.0.0 test
> jest tests/alerta.test.js

PASS tests/alerta.test.js
  ✓ alerta critico se nivel maior que 80 (2 ms)
  ✓ alerta alto se nivel entre 51 e 80 (1 ms)
  ✓ alerta moderado se nivel até 50 (1 ms)

Test Suites: 1 passed, 1 total
Tests: 3 passed, 3 total
Snapshots: 0 total
Time: 0.28 s, estimated 1 s
Ran all test suites matching tests/alerta.test.js.

C:\Users\joaov\OneDrive\Área de Trabalho\Faculdade\DSM_4S\IEC\1 - aula github\inpe-alertas>
```

Exercicio 07



The screenshot shows the VS Code editor with a file named `integracao.test.js` open. The code defines a mock `api` object with an `enviar` method and a single Jest test. The terminal at the bottom shows the command `npx jest tests/integracao.test.js` being executed, resulting in one passed test. The status bar at the bottom indicates the file is `feature/seguranca` and the user is `jrb`.

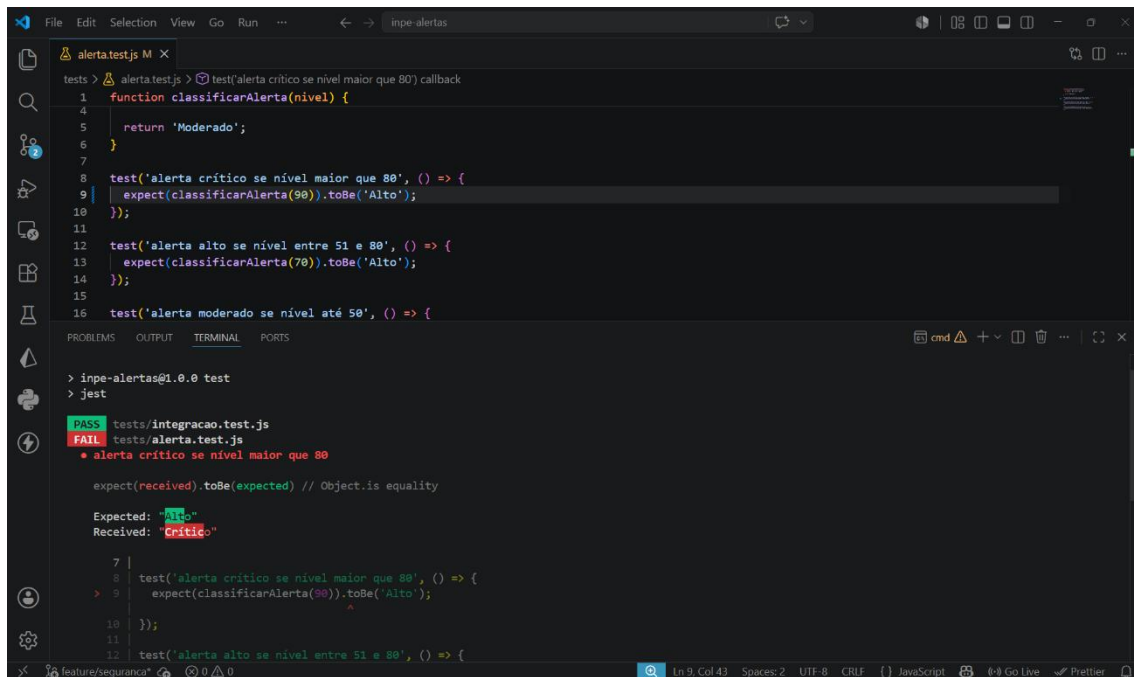
```
tests > mock.test.js > ...
1 const api = {
2   enviar: jest.fn(() => 'Simulado!'),
3 };
4
5 test('simulação de envio', () => {
6   const resposta = api.enviar();
7
8   expect(resposta).toBe('Simulado!');
9 });
```

```
C:\Users\joaov\OneDrive\Área de Trabalho\Faculdade\DSM_4S\IEC\1 - aula github\inpe-alertas>npx jest tests/integracao.test.js
PASS tests/integracao.test.js
  ✓ classificação + notificação (2 ms)

Test Suites: 1 passed, 1 total
Tests: 1 passed, 1 total
Snapshots: 0 total
Time: 0.208 s, estimated 1 s
Ran all test suites matching tests/integracao.test.js.

C:\Users\joaov\OneDrive\Área de Trabalho\Faculdade\DSM_4S\IEC\1 - aula github\inpe-alertas>
```

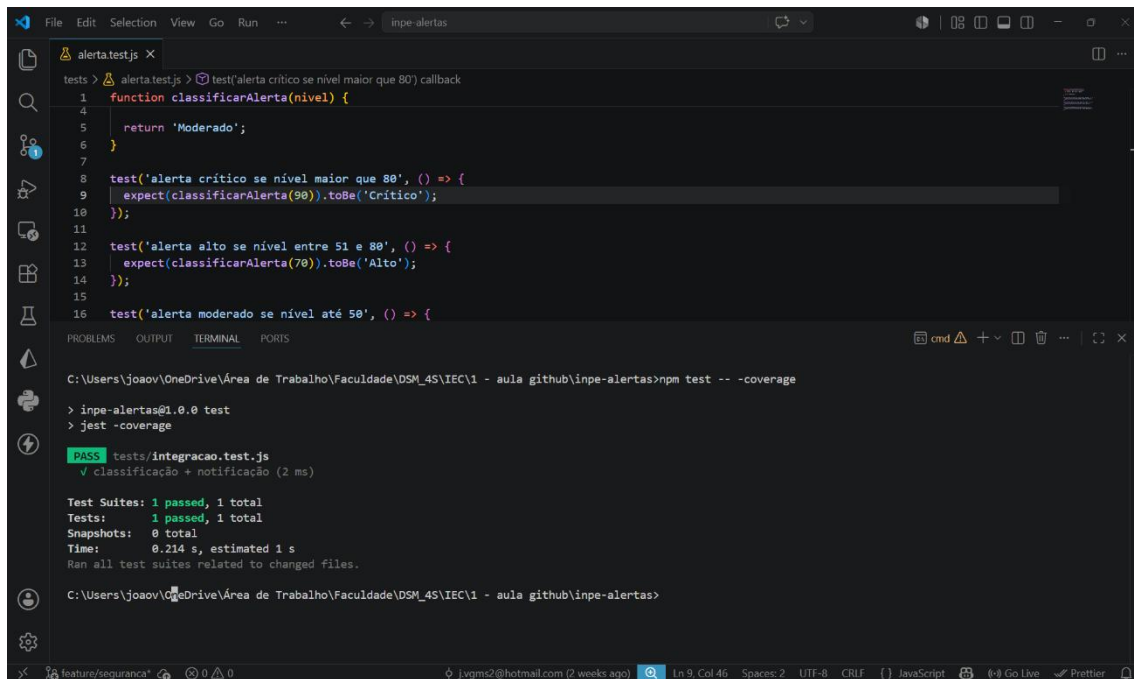
Exercicio 08



The screenshot shows a VS Code editor with a file named `alerta.test.js`. The code defines a `classificarAlerta(nivel)` function and three Jest tests. The first test, `test('alerta critico se nivel maior que 80', () => { expect(classificarAlerta(90)).toBe('Alto'); });`, is failing. The terminal output shows the test results: `PASS tests/integracao.test.js` and `FAIL tests/alerta.test.js`. The failure message is `* alerta critico se nivel maior que 80`. The error details show `expect(received).toBe(expected) // Object.is equality` with `Expected: "Alto"` and `Received: "Crítico"`. The code in the editor is as follows:

```
1 function classificarAlerta(nivel) {
4
5   return 'Moderado';
6 }
7
8 test('alerta critico se nivel maior que 80', () => {
9   expect(classificarAlerta(90)).toBe('Alto');
10 });
11
12 test('alerta alto se nivel entre 51 e 80', () => {
13   expect(classificarAlerta(70)).toBe('Alto');
14 });
15
16 test('alerta moderado se nivel até 50', () => {
```

Exercicio 09



The screenshot shows the same VS Code editor with the `alerta.test.js` file. The terminal output shows the test results for `inpe-alertas@1.0.0 test` and `jest -coverage`. The test results are: `PASS tests/integracao.test.js` and `✓ classificação + notificação (2 ms)`. The test suites and tests are all passed. The error details show `Expected: "Alto"` and `Received: "Crítico"`. The code in the editor is as follows:

```
1 function classificarAlerta(nivel) {
4
5   return 'Moderado';
6 }
7
8 test('alerta critico se nivel maior que 80', () => {
9   expect(classificarAlerta(90)).toBe('Crítico');
10 });
11
12 test('alerta alto se nivel entre 51 e 80', () => {
13   expect(classificarAlerta(70)).toBe('Alto');
14 });
15
16 test('alerta moderado se nivel até 50', () => {
```